

The transition to a sustainable economy is not sustainable: an ecological macroeconomic analysis of climate policy, the circular economy, and global inequality

Oriol Vallès Codina, José Bruno R. T. Fevereiro, Annina Kaltenbrunner, Effie Kesidou, Andrea Genovese, Marco Veronese Passarella¹

*ABSTRACT***Introduction**

The transition to sustainability is nowadays understood to go beyond the energy transition, requiring the adoption of circular-economy (CE) strategies that seek to reduce material throughput, as well as emissions, increasing efficiency in resource use (Geissdoerfer et al., 2017; Korhonen et al., 2018). In recent years, the global policy landscape has begun to shift from high-level "roadmaps" to legally binding regulations and sector-specific enforcement (Barrie et al. 2024). Results from most academic studies, conducting ex-ante analysis, have projected positive environmental and socio-economic impacts of the adoption of CE strategies, specially, at the national or regional level (Agullar-Hernandez et al., 2018, and Fevereiro et al., 2025).

Yet, the effects of circular transition cannot be assessed only in the country where policy is legislated. This is because supply chains, trade relations, and financial hierarchies connect countries and regions asymmetrically. This implies that policies designed for green production and consumption in advanced economies may reallocate costs, demand pressures, and ecological burdens across countries rather than resolve them globally (Wiedmann et al., 2015; Dorninger et al., 2021). Moreover, recent research suggests that circularity and decarbonisation do not necessarily move together, since circular initiatives can generate energy-intensive processes, upstream emission increases, or burden-shifting across supply chains when they are not coordinated with broader net-zero strategies (Arranz et al., 2026).

Notwithstanding these issues, most CE research focuses on technical efficiency and firm-level practices, overlooking the macroeconomic and global structures through which environmental and socio-economic effects are distributed (Kirchherr et al., 2017; Geissdoerfer et al., 2017). Less attention is paid to the structural relations through which environmental benefits and

¹ **Affiliations:**

Oriol Vallès Codina (Net Zero Industrial Policy Lab, Johns Hopkins University, 3400 N Charles St. Baltimore, MD 21218)

José Bruno R. T. Fevereiro (Banco Nacional de Desenvolvimento Econômico e Social, Brazilian Development Bank)

Annina Kaltenbrunner (Leeds University Business School)

Effie Kesidou (University of Bath)

Andrea Genovese

Marco Veronese Passarella (Università L'Aquila, Leeds University Business School)

socio-economic costs are distributed internationally, and to the fact that different CE channels do not have the same macroeconomic or ecological effects. However, circular transitions occur within a global economy that is characterised by unequal ecological exchange, persistent structural asymmetries, and monetary and financial constraints (Hornborg, 2009; Hickel et al., 2022). Here, therefore, we argue that the question is not whether CE practices, in isolation, reduce emissions or material extraction, but (i) under what conditions they do so, (ii) for whom, and (iii) through which demand and production channels, and (iv) with what macroeconomic and distributive consequences across regions.

Our paper addresses this gap through a two-region ecological macroeconomic model combining multi-regional input–output structure and stock–flow consistent dynamics, calibrated for the European Union and the Rest of the World using EXIOBASE 2011 data and 54 industries covering eight basic materials — food, energy, wood, plastics, pulp, glass, cement, and construction — each disaggregated into a primary (virgin, resource-intensive) and a secondary (resource-efficient) production sector. We simulate 14 circular economy transition scenarios, each targeting a different primary-to-secondary substitution pathway and varying systematically by demand channel (final demand, intermediate production, investment) and institutional actor (households, government, firms).

Model Overview

In this section we provide a conceptual overview of the main characteristics of the different modules of the model². The analysis relies on a dynamic, ecological, multi-regional input-output, stock-flow consistent (IO-SFC) model calibrated on empirical Input-Output data (Exiobase 3). The model is designed to reproduce the evolution of an interconnected macro-financial-ecological system and to test the implications of alternative circular-economy (CE) trajectories.

Several features are worth highlighting.

1. The model operates in discrete time and captures how system-level behaviour emerges from the interaction of its components, laying emphasis on financial-real interactions.
2. It enforces stock-flow consistency across all sectors, ensuring that every flow has a corresponding change in a stock and that monetary, real, and physical balances hold by construction.
3. The production side is represented through a two-region input-output structure, so all inter-industry linkages - within each region and across regions - are explicitly accounted for.
4. The framework is fully open-economy: the world is divided into two regions (the European Union and the rest of the world), connected through goods trade and cross-border portfolio flows.

² The full description of the equations of the model is presented in appendix 1.

5. Ecological relations are endogenised. Emissions, resource extraction, and waste generation are linked directly to production technologies and material flows.
6. Finally, the numerical parameters of the model are anchored in observed data, either drawn directly from existing time series (e.g. technical coefficients) or calibrated to replicate the prevailing levels of key macroeconomic aggregates.

In recent years, several authors (e.g., Hardt and O'Neill 2017; Bimpizas-Pinis et al., 2023; Fevereiro et al., 2025) have identified the combination of input-output (IO) analysis and stock-flow consistent (SFC) modelling as one of the most promising approach for developing models assessing the economy-ecology nexus, including Circular Economy (CE) transition scenarios.

As is well known, IO models are analytical tools used to represent and quantify the interdependencies between different industries of a capitalist economy (e.g., Leontief 1936, 1941). More specifically, IO models illustrate how changes in one industry, such as increased production or consumption, affect other industries through a system of interconnected inputs and outputs, providing insights into the overall economic impact of alternative shocks and policies.

On the other hand, SFC models can be considered a specific class of 'system dynamics' tools that replicate the dynamics of a financially sophisticated economy (e.g., Godley and Lavoie 2007, Caverzasi and Godin 2015, Nikiforos and Zezza 2017). In the last decade, SFC models have gained traction in ecological macroeconomics too, due to their ability to integrate consistently and comprehensively the flows and stocks of the real economy, the financial sector, and the ecosystem (Dafermos 2017, 2018; Carnevali et al., 2019, 2020, 2023).

Due to the complexity of an integrated approach, only a limited number of studies have so far incorporated explicitly the IO structure of the economy into a SFC dynamic framework (e.g. Berg et al., 2015; Valdecantos and Valentini, 2017; Di Domenico et al., 2024; Thomsen et al., 2025; Jackson and Jackson, 2025; Pettena and Raberto, 2025; Veronese Passarella, 2025). Nevertheless, some progress has been achieved in recent years. In particular, Veronese Passarella (2023) has extended a standard aggregative SFC model by introducing both a vertical disaggregation (across institutional sectors) and a horizontal disaggregation (across production industries), and has used this framework to explore simple circular economy scenarios. Subsequent contributions have further developed this line of research by broadening the scope of application and refining the treatment of inter-industry and cross-border linkages. For instance, Fevereiro et al. (2023) have expanded on Veronese Passarella (2023)'s analysis by applying it to a two-area economy. This extension explicitly considers the effects of international trade, supply-chain interdependencies, cross-border portfolio investments, and exchange rate fluctuations.

Formally, the model consists of accounting identities and behavioural equations relating industries and institutional sectors. These are organised into three blocks: the economic-financial block, the social block, and the ecological block.

Economic and Financial Block

Each region comprises five institutional sectors: households, private firms, government, commercial banks, and the central bank. The institutional structure of the economy is identical in the two regions, and in the baseline no trade or financial restrictions are imposed.

Households earn wages and capital income (distributed profits, interest, and capital gains) and allocate their disposable income across consumption and asset holdings. Consumption is a fixed function of disposable income and wealth, while portfolio allocation follows a Tobin-type mechanism based on relative returns and liquidity preferences. Households may take out personal loans to finance durable consumption or expenditure exceeding current income.

Contrary to a typical SFC model, private firms are disaggregated into 54 productive sectors based on IO data. Unless explicitly affected by change in one of the scenarios tested, firms operate under assumption fixed input proportions (Leontief technology) and constant returns to scale assumption. Industries produce homogeneous outputs and set prices via mark-ups over unit costs, which include intermediate inputs, labour costs and the depreciation of the capital stock. Firms adjust their capital stock toward a target capital–output ratio; this generates investment demand even when output is constant. Investment is financed through retained earnings, bank credit, and the issuance of shares purchased by domestic and foreign households.

Government consumption expenditure - both consumption and public investment - is exogenous. Revenues accrue from income taxes, VAT, import duties, and central-bank profits. Budget deficits are financed through the issuance of government bills. Central banks accommodate the demand for currency and purchase the portion of government debt not absorbed by the private sector. The policy rate set by the central bank determines the interest rates on loans, deposits, and government bills through sector-specific mark-ups.

Commercial banks supply credit elastically: firms receive all the financing they require for production and investment. Deposits exceeding the volume of loans are placed in government bills; if loans exceed deposits, banks borrow from the central bank.

In the CE scenarios, part of the waste stream is diverted to a recycling industry that produces secondary inputs. This is implemented through changes in the relevant input–output coefficients.

Social Block

Although households are modelled as an aggregate sector, the framework tracks the distribution of income and wealth between workers and rentiers, both before and after taxes. This allows the model to assess distributional consequences of alternative CE trajectories.

Employment is determined by industry-specific labour demand. Population dynamics, and thus the size of the labour force, follow exogenous demographic growth and net immigration inflows. Cross-regional migration reacts to population size, unemployment differentials, and wage gaps.

Ecological Block

The ecological sub-model follows recent advances in ecological SFC modelling. Waste generation is tied to industrial production through sector-specific coefficients. In the baseline, waste flows are processed by traditional waste-management industries; in CE scenarios, new recycling/reuse channels are introduced.

Industrial CO₂ emissions depend on the use of non-renewable energy inputs. Emissions are determined by energy-intensity coefficients, the share of non-renewables in each industry's energy mix, and a uniform emissions factor. Accumulated emissions contribute to atmospheric concentrations, which feed into a simplified climate-response module linking cumulative emissions to temperature change.

Material extraction and energy use reflect both new production and the recycling of socio-economic stocks. Renewable energy is replenished periodically, whereas non-renewables are depleted. The size of material and energy reserves evolves according to extraction and conversion flows.

Data Collection and Calibration

Model simulation is carried out in R, using EXIOBASE 3 v 3.8.2, with data calibrated for the baseline year of 2011. The original 164-sector, 49-region classification is aggregated into two regions (EU and rest of world) and 54 sectors, among which 13 sectors specifically produce goods based on re-processed material inputs³.

From EXIOBASE data we derive the technical coefficients, labour productivity, real wages, sectoral employment (including gender composition), final demand structure, emissions coefficients, and material-use data required to initialise the technological and environmental parameters. In turn, macro aggregates such as GDP, final demand components, trade flows, and gross output serve as calibration targets.

Remaining parameters are assigned through an evolutionary random-search routine. The algorithm perturbs an initial set of parameter values, evaluates the resulting steady-state values of key variables against their targets, and accepts parameter updates only when the goodness-of-fit improves. This process continues until a satisfactory baseline is reached, subject to constraints ensuring economically meaningful outcomes⁴. (e.g., propensity to consume out of class of income-hierarchy conditions).

This calibration phase must also stabilize the system so that, at the moment policy shocks are introduced, the model is near a steady configuration: GDP, trade balances, public deficits, sectoral balances, and stock-flow norms must remain stable over time. The resulting calibration yields a good but improvable match to observed data, with consumption slightly overestimated in the EU and investment underestimated in both regions.

³ A complete list of the sectors is provided in Appendix 2

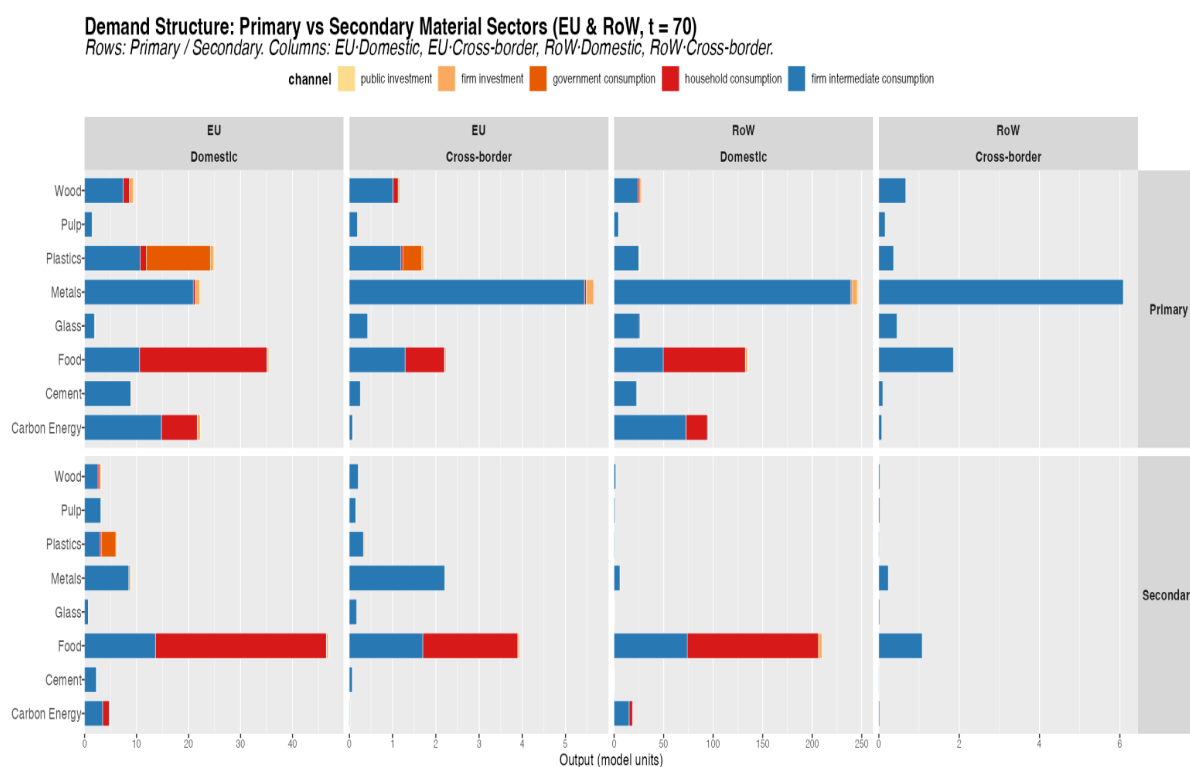
⁴ For example, following post-keynesian theory, a constraint is introduced to ensure parameters values for the propensity to consume out of wages is higher than out of profit and, in turn, out of wealth.

Descriptive Statistics: the Multi-Regional Input-Output Structure of Demand

This section documents the multi-regional input-output (MRIO) structure of demand for eight key material pairs in the EU and Rest of World (RoW), as calibrated in the SOCIO-ECO-MRIO-SFC model at its baseline steady state at $t = 70$. The figures presented here are descriptive statistics derived analytically from the IO accounting framework without any simulations.

Figure 1 decomposes gross output for all eight material pairs by institutional demand channel (household consumption, government consumption, firm investment, public investment, and firm intermediate consumption) across four regional flow dimensions: EU domestic production for EU demand, EU cross-border imports from RoW, RoW domestic production, and RoW cross-border exports to EU.

Domestic/Cross-Border, Primary/Secondary Composition



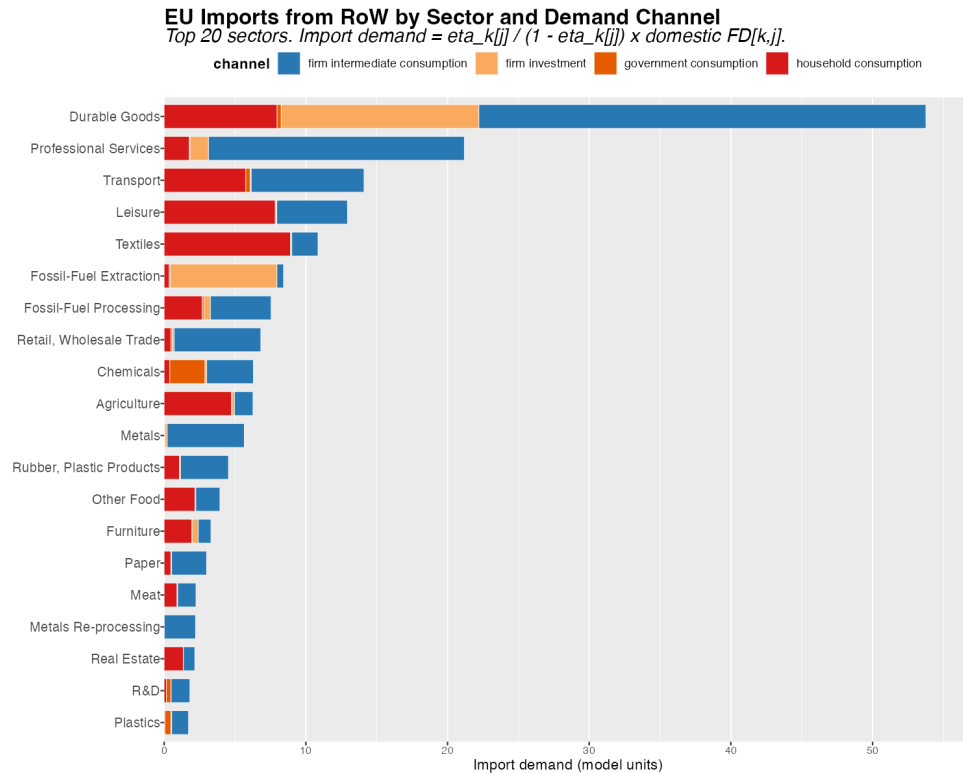
Note: Gross output decomposition for primary and secondary material sectors across EU and RoW. Columns: EU Domestic, EU Cross-border, RoW Domestic, RoW Cross-border demand. Rows: Primary vs Secondary. Bars stacked by institutional demand channel.

Some important patterns are worth highlighting.

Key patterns:

- **Industrial materials (cement, glass, metals, pulp)** are overwhelmingly driven by firm intermediate consumption, which accounts for 93–96% of gross output in the EU. Final demand channels are negligible. This has direct implications for CE policy design: demand-side instruments (eco-labelling, green public procurement) operate on a structurally marginal share of these markets.
- **Food** (primary: Meat; secondary: Other Food) stands at the opposite extreme: 65% of EU domestic output serves household consumption. This is the only material sector where a consumer-facing intervention can engage the dominant demand channel.
- **Carbon Energy** (primary: Fossil-Fuel Electricity; secondary: Renewable Electricity) is structurally intermediate, with approximately 34% final demand (predominantly household electricity consumption) and 66% intermediate. Industrial energy procurement is the primary driver. Secondary output (renewables) is substantially more intermediate-heavy than primary (fossil fuels), reflecting that renewable electricity is primarily procured by industrial users, while fossil-fuel electricity has a larger direct household retail component.
- **Plastics** present an exceptional structure: while approximately 55% of output serves final demand, this component is dominated by government consumption rather than households. Private investment and household channels are minor. This structural feature of the plastics market makes government procurement a first-order policy lever.
- **Primary vs Secondary gap** Secondary sectors (recycled/renewable alternatives) are systematically smaller than their primary counterparts in absolute terms. The compositional mix of secondary output also differs — in several materials, secondary sectors are more intermediate-heavy, suggesting industrial early adopters are leading the transition ahead of final consumers.
- The **EU domestic vs cross-border column comparison** reveals that cross-border (import) demand tends to be more intermediate-heavy, particularly for manufactured materials (plastics, metals). This reflects the role of global value chains in sourcing industrial inputs from RoW suppliers.

Figure 3: EU Import Structure: EU Demand for RoW Goods



Note: EU import demand from RoW decomposed by sector and demand channel. Includes firm intermediate consumption, household consumption, government consumption, and firm investment. Top 20 sectors by total import demand.

Figure 3 maps the cross-border component of EU demand — imports from RoW — broken down by institutional channel, including both final demand channels (households, government, fixed investment) and firms’ circulating investment.

Key Patterns

- **Firm intermediate consumption dominates EU import demand across most high-import sectors.** For energy and manufactured goods, the bulk of cross-border purchases are industrial inputs, not consumer goods. This implies that CE policies targeting intermediate production substitution have the largest potential to reduce import dependency alongside domestic primary material use.
- **Fossil fuels and carbon energy rank among the highest-import sectors**, with a pronounced intermediate consumption component reflecting the EU’s structural dependence on imported fossil-fuel inputs for industrial production processes.
- **Government and household consumption import shares are non-trivial for food and certain consumer goods**, identifying channels where procurement policy and consumption-side interventions could reduce cross-border material footprints.
- **Ranking of sectors by total import demand directly anticipates which CE transitions will generate the largest trade balance improvements in the dynamic simulations:** scenarios

reducing demand in high-import primary sectors will produce the largest improvements in the EU current account balance.

Leontief Decomposition

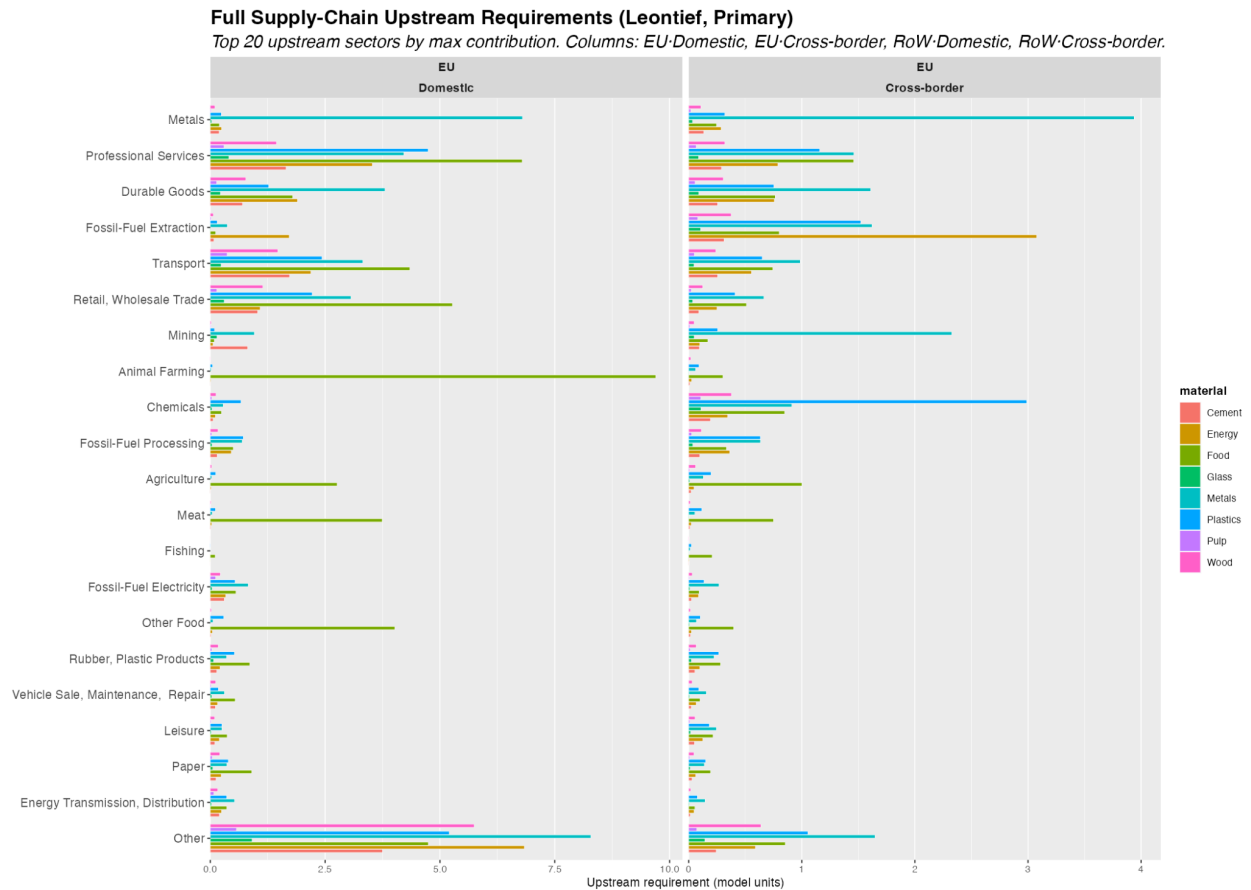


Figure above presents the column view of the Leontief upstream decomposition: $(L - I)_{kt} \cdot x_t$, showing which sectors k are required as inputs — across all orders of IO interdependence — to sustain the gross output x_t of each material sector t . This is the full supply-chain view, encompassing not just direct inputs (captured by the A matrix) but all indirect upstream requirements propagated through the IO system.

Key Patterns

- Animal Farming is the dominant upstream input for Food (primary: Meat). In the EU domestic column (Figure 7), Animal Farming ranks first across the Food material — the full supply-chain view makes explicit what is obvious from the production structure: meat processing draws on livestock rearing across all IO orders. Notably, in the EU cross-border column, Fossil-Fuel Electricity and Metals appear as significant cross-regional upstream requirements, connecting food production to global energy and industrial supply chains.
- Fossil-fuel extraction and processing are the dominant upstream inputs for primary Carbon Energy (Fossil-Fuel Electricity). This connection — invisible in direct intermediate demand statistics — becomes salient in the full supply-chain view. Primary fossil-fuel electricity generation draws heavily on fossil-fuel extraction (EU domestic) and fossil-fuel processing as upstream requirements across all IO orders. This validates the model's treatment of carbon energy scenarios as a pivot point for cross-sectoral decarbonisation.
- Metals shows the largest aggregate Leontief contribution across all material sectors, driven by the economy-wide role of metals as an input to machinery, infrastructure, and energy systems. Crucially, comparing the primary (Figure 7) and secondary (Figure 8) columns for Metals reveals the precise mechanism behind the emission rebound quantified in Section 7. The structural factor acts symmetrically: shifting demand from primary (sector 26) to secondary (sector 27) reduces upstream requirements for sector 26 by unit (,) and increases requirements for sector 27 by unit (,). Because Metals Re-processing is itself more emission-intensive than primary Metals (vs), the increase in sector 27 requirements costs more in emissions (+443) than the reduction in sector 26 saves (−293). A secondary contribution comes from Fossil-Fuel Electricity (sector 31), which secondary metals requires more intensively (,), adding +74 to the rebound due to its very high emission intensity (,). The savings from reduced Mining and Fossil-Fuel Extraction — inputs that primary metals needs for iron ore and coal — partially offset these but are insufficient to reverse the net sign.
- Primary vs secondary supply-chain divergence: Comparing Figures 7 and 8, secondary sectors generally exhibit different upstream structures relative to their primary counterparts. Secondary Carbon Energy (Renewable Electricity) has substantially lower fossil-fuel upstream requirements, confirming the ecological rationale for the transition. For Metals, the divergence works in the opposite direction for emissions — secondary Metals Re-processing requires more of itself and more fossil-fuel electricity — producing the rebound documented in Section 7.
- The EU–RoW cross-border column reveals which upstream requirements are supplied cross-regionally. For Carbon Energy and Metals, significant upstream requirements originate from RoW sectors, underscoring the importance of global supply chain dynamics in evaluating the full ecological impact of EU CE transitions.
- Sector ordering (largest aggregate contribution at top) reveals that energy-related sectors, Animal Farming (for Food), construction, and financial/business services constitute the dominant upstream inputs across materials — reflecting the economy-wide energy, agricultural, and services intensity of all production processes.

Circular Economy Policy Portfolio Effectiveness

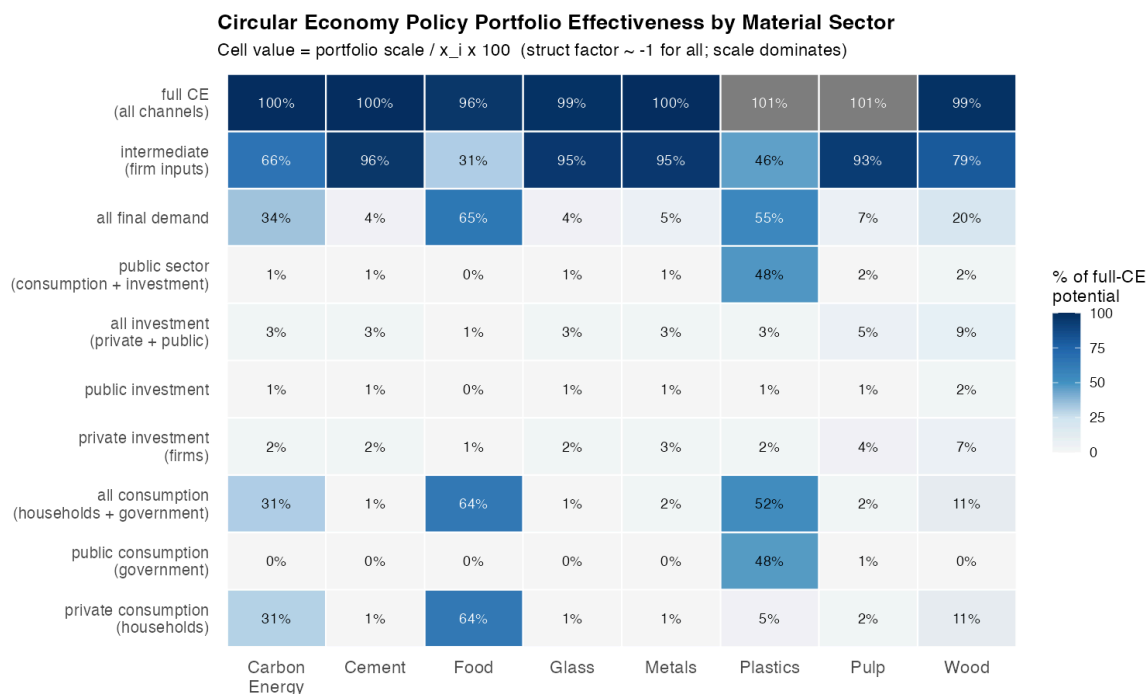


Figure 4: CE policy portfolio effectiveness by material sector. Cell values report the portfolio scale as a percentage of gross output x_i , which — given the structural factor $(l_{i2} - l_{i1})$ approximately -1 for all materials — equals the percentage of full-CE potential captured by that portfolio at substitution intensity $\rho = 0.2$.

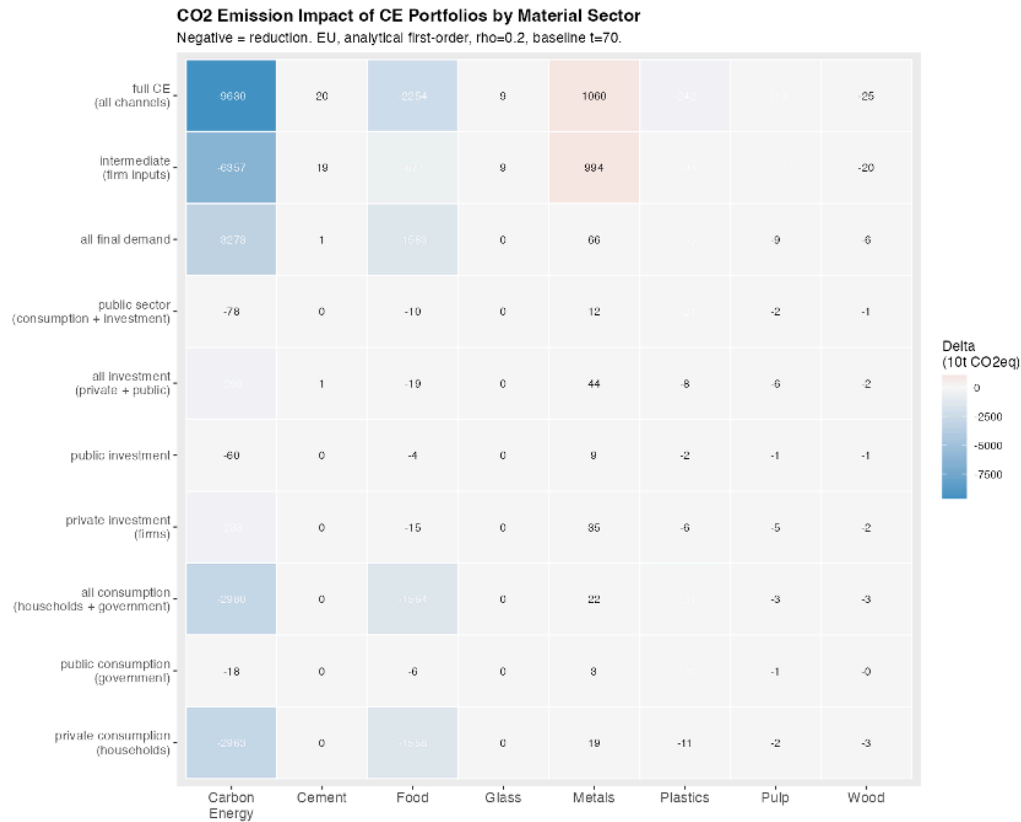
Figure 4 systematically compares the effectiveness of seven CE policy portfolios regarding the demand channels across eight material sectors, providing the key analytical bridge between the IO structure and the simulation outcomes. Because the structural factor is approximately constant across all material pairs, differences in simulation outcomes will be driven almost entirely by differences in the structure of the demand under intervention, not by variation in the IO multiplier. The heatmap therefore anticipates the ordering of cumulative material savings, emission reductions, and macroeconomic effects across scenarios.

Key Patterns

- Intermediate production substitution dominates for all industrial materials.** For cement, metals, glass, and pulp, the intermediate channel alone captures 93–96% of the full-CE benchmark. Any final demand portfolio — even combining all four institutional components — recovers less than 8% of potential. This is not a marginal advantage of production-side policy; it is a categorical difference in order of magnitude.
- Plastics:** government procurement equals intermediate substitution. For plastics, the government consumption channel alone (47.5%) is comparable to the intermediate channel (45.9%) and far exceeds household consumption (4.6%). This formal equivalence — unique among the eight

materials — reflects the institutional dominance of government procurement in the EU plastics market and makes circular public procurement a first-order policy instrument alongside production-side regulation.

- **Food:** the household channel leads. At 63.8%, household consumption is the dominant CE lever for food — the only material where this is true. Even so, the intermediate channel adds a complementary 31%, pointing to the importance of food processing and catering supply chains.
- **Carbon Energy: firm intermediate dominates over household behaviour.** Despite public discourse focusing on household energy transitions, household consumption captures only 30.6% of the full-CE potential for carbon energy. The intermediate channel captures 66.1%. Industrial energy procurement — not consumer behaviour — is the primary driver of fossil-fuel electricity demand.
- Carbon energy transitions generate by far the largest absolute emission and material reductions. The intermediate channel alone yields $\Delta \text{CO}_2 \approx -6,357$ (10t CO₂eq) at $\rho = 0.2$, and the full-CE portfolio yields $-9,639$. This reflects both the large portfolio scale (intermediate channel dominates at 66%) and the high emission intensity of fossil-fuel electricity generation.
- Structural rebound is a real constraint for industrial materials. For glass, cement, and metals, the secondary sector's upstream supply chain is currently more emission-intensive than the primary sector's. For Metals, the Leontief decomposition identifies two specific channels: (i) Metals Re-processing is itself more emission-intensive than primary Metals ($\epsilon_{27} > \epsilon_{26}$), so the symmetric $\approx +1$ increase in sector 27 requirements outweighs the ≈ -1 reduction in sector 26; and (ii) secondary metals requires more fossil-fuel electricity than primary metals ($L_{31,27} > L_{31,26}$). Both channels must be addressed — through decarbonisation of the electricity grid and cleaner reprocessing technology — before CE transitions in metals can deliver net emission reductions.



Scenario Design

By combining ecological and input-output extensions with a macro-financial open-economy framework, the model allows us to analyse circular-economy scenarios as structurally differentiated demand transition regimes rather than isolated technical shocks. Specifically, we compare how sector-level changes in final demand, intermediate production, and sectoral expansion affect value added, employment, income distribution, fiscal and current-account balances, emissions, and material extraction domestically and across borders. In doing so, we examine how circular-economy policies generate transnational socio-ecological trade-offs and cross-regional macro-financial asymmetries.

We apply the model to analyse fourteen circular economy scenarios, each redirecting a share of primary material demand toward secondary alternatives across specific demand channels identified in the MRIO structure

Scenario Analysis

The simulations reveal three findings.

First, there is a single CE transition regime but a set of demand channel- and sector-specific outcomes, as the magnitude of ecological and macroeconomic effects is determined by the size of the demand channel targeted. Household food transitions produce the largest final-demand employment contraction and meaningful material savings, reflecting one third of household share in food demand. Household energy transitions deliver strong emission reductions disproportionate to their one-third channel share, amplified by the 3.3:1 emission intensity differential between fossil and renewable electricity. In contrast, government procurement constitutes a key policy lever for plastics due to its unusually large demand share.

Second, intermediate production restructuring delivers stronger ecological gains but at a structural cost: because secondary sectors require fewer intermediate inputs per unit of output than their primary counterparts, the very substitution that reduces material intensity also contracts overall output and compresses the wage share. Energy restructuring is the highest-leverage scenario in emissions and primary material extraction (reflecting the combination of large intermediate demand and a high emission intensity differential between fossil and renewable energy, more land-intensive renewable infrastructure increases land use. Metals is the only production scenario to expand output and employment while delivering the largest material savings, yet generates a structural rebound effect on carbon emissions through the higher fossil-electricity intensity of secondary steel production.

Across the full set of scenarios, the results show a key structural pattern –only visible in the MRIO framework– that circular-economy transitions are characterised by transnational socio-ecological trade-offs as the EU achieves ecological gains does so partly by displacing demand from the Rest of the World. Ecological improvements, macroeconomic performance, and distributional outcomes do not move together automatically. In particular, high-leverage transitions, such as energy and metals, generate asymmetric cross-regional effects, including shifts in employment, external balances, and fiscal positions between the EU and the rest of the world.

Four distinct cross-border transmission patterns emerge.

- Symmetric material contraction — the most common pattern across intermediate scenarios for wood, pulp, glass, and cement — reduces material throughput, output, and employment modestly in both regions simultaneously, as EU restructuring contracts global intermediate demand rather than simply shifting production abroad.
- Competitive displacement occurs in the metals scenario, where the lower intermediate input requirements of secondary steel production generate a net EU expansion in output and employment and a current-account improvement, while RoW primary metal producers contract.

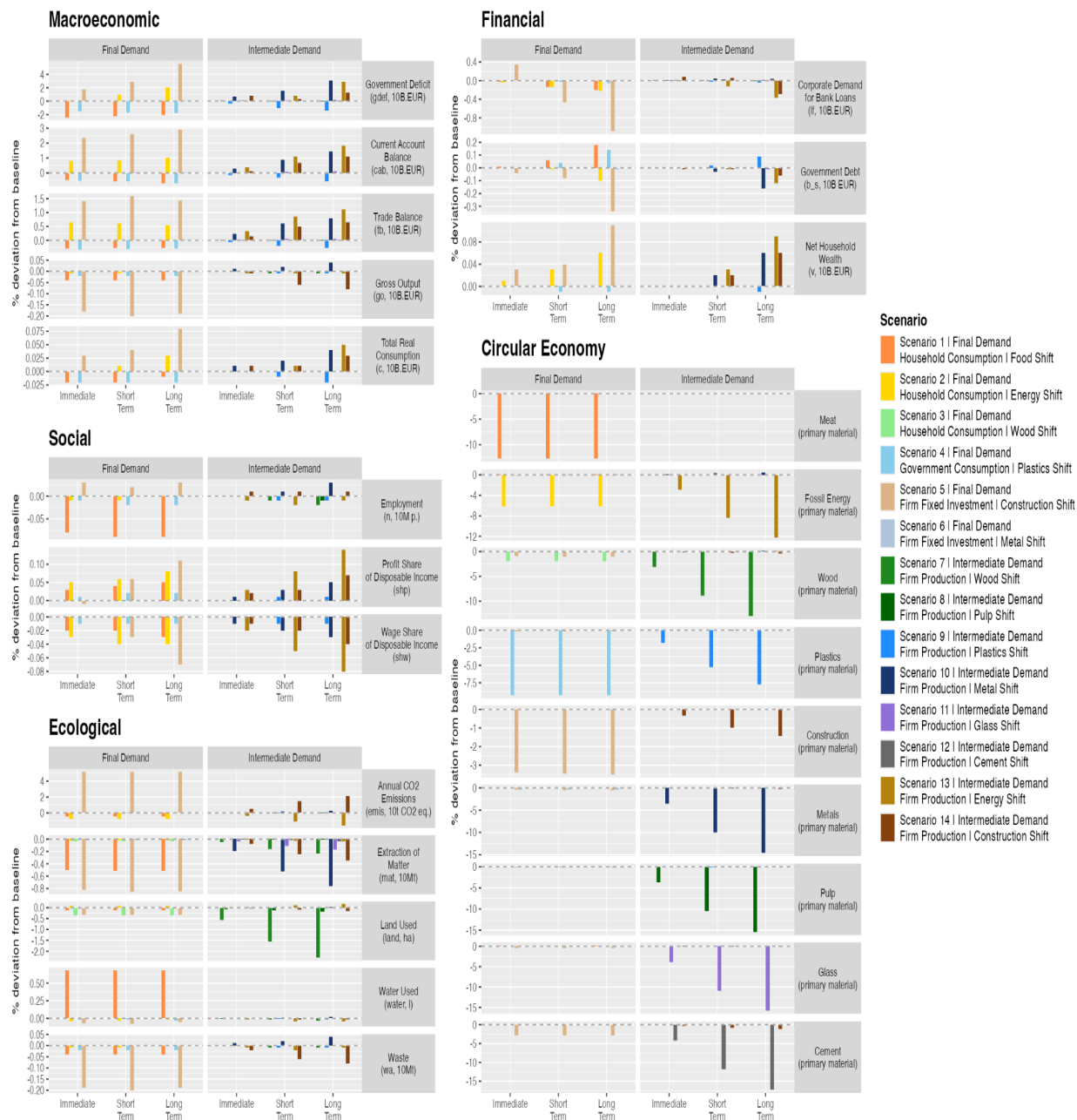
- Fossil-import collapse — the most asymmetric pattern — occurs in the energy scenario: the EU gains a current-account dividend and falling public debt as fossil imports collapse, while RoW fossil exporters face worsening government deficits and rising public debt, alongside the sharpest compression of the wage share within the EU.
- Finally, partial production leakage in plastics and food runs in the opposite direction: because secondary sectors carry higher import content, EU substitution generates small positive RoW employment effects while worsening the EU current account, and in the plastics case produces a measurable cross-border emission leakage.

These transnational trade-offs are structurally embedded in the global production system and invisible to single-region frameworks.

Scenario Registry

Circular Economy Intervention Comparison | Domestic Effects | EU

Shock Parameter $p = 0.2$ | $p = 0.05$ for Construction (Sc5 & Sc14) | All flows except household net wealth, government debt, and loans | % deviation from baseline



Circular Economy Intervention Comparison | Cross-Border Effects | RoW

Shock Parameter $\rho = 0.2$ | $\rho = 0.05$ for Construction (Sc5 & Sc14) | All flows except household net wealth, government debt, and loans | % deviation from baseline



(cross-border) effects.

The comparison figures present all 14 scenarios simultaneously across three time horizons: Immediate ($t+1$), Short-Term ($t+7$), and Long-Term ($t+20$). The integrated figure (08/09) organises variables into five categories across two columns: Macroeconomic, Social, and Ecological on the left; Financial and Circular Economy sector outputs on the right. Separate figures are produced for EU (domestic) and RoW

Scenario Discussion

The Portfolio Scale Principle

The most robust finding across all 14 scenarios is that the rank ordering of outcomes is predicted by the absolute scale of the targeted demand channel, not by material-specific multiplier variation or sectoral intensity alone. The structural Leontief factor ($l_{12} - l_{11}$) is approximately -1 across all material sectors, meaning that primary-to-secondary substitution consistently contracts aggregate demand in proportion to channel size. What varies decisively across scenarios is the size of the demand portfolio the policy instrument engages. This generates a clear ranking:

- **Food (Sc1)** produces the largest final-demand employment contraction ($Z1\ n = -0.09\%$ LT) because household food demand (23.7 B EUR) is the largest household material channel. It also drives the largest primary material extraction reduction among FD scenarios ($Z1\ x_mat = -0.30\%$), despite modest emission savings (-0.42%), because the food sector is more material-intensive than emissions-intensive.
- **Household energy (Sc2)** produces strong emission and primary material savings ($Z1\ emis = -0.78\%$, $Z1\ x_mat = -0.64\%$) disproportionate to its channel size (6.9 B EUR) because the 3.3:1 fossil-to-renewable intensity differential amplifies the scale effect on both emission and primary energy extraction.
- **Government plastics (Sc4)** has disproportionate leverage among FD scenarios because the government procurement channel (11.9 B EUR) rivals the intermediate channel in absolute scale — unique among all materials. However, the material effect is ambiguous: EU total material use barely changes ($+0.01\%$) while primary extraction falls only -0.04% , reflecting small primary–secondary intensity gaps for plastics.
- **Energy production (Sc13)** dominates all production scenarios: $Z1\ x_mat = -1.27\%$ LT (the largest primary material extraction reduction in the entire set), driven by the combination of the largest intermediate channel (14.8 B EUR) and the 3.3:1 intensity differential. Total material use ($Z1\ mat = -0.04\%$) moves little because energy restructuring primarily displaces fossil fuel use (captured in x_mat) rather than bulk material throughput.
- **Metals production (Sc10)** produces the largest total material savings ($Z1\ mat = -0.76\%$ LT) because intermediate metal demand (20.9 B EUR) is the largest single intermediate material flow in the scenario set. The A-matrix row-sum of secondary metals (0.238) is less than half that of primary metals (0.545), amplifying material throughput reductions across the supply chain.

The simulation ranking therefore validates the demand channel decomposition: before running the model, relative outcome magnitudes can be predicted from channel scale $\times$ intensity differential with reasonable accuracy.

Two Transition Regimes

The 14 scenarios separate into two regimes distinguished by ecological leverage and macroeconomic direction.

Final-demand reallocation (Scenarios 1–6) is not a uniformly low-leverage regime. This characterisation — common in single-region IO analysis — understates the scale of several channels once bilateral MRIO demand flows are included. Scenario 1 (Food, HH) produces the largest employment contraction among FD scenarios ($Z1\ n = -0.09\%$ LT) because food demand (23.7 B EUR) is the dominant household material channel. Scenario 2 (Energy, HH) generates ecological gains ($emis = -0.78\%$, $x_mat = -0.64\%$) disproportionate to its channel size because the 3.3:1 intensity differential amplifies scale into emission and primary material extraction reduction. Scenario 4 (Gov Plastics) is structurally distinctive: government procurement accounts for 48.9% of plastics demand (11.9 B EUR), making public procurement circularity mandates a genuine first-order CE instrument for this material. Scenario 5 (Construction FD) activates a large fixed-investment channel (88.7 B EUR of total construction demand) and is the only FD scenario producing a positive employment effect ($+0.03\%$ LT) because renovation is

more labour-intensive than new build. Final-demand CE instruments engage channels that are structurally primary for food, construction, and plastics, and cannot be treated as merely symbolic complements to production-side policy.

Intermediate production restructuring (Scenarios 7–13) delivers substantially higher ecological gains at the cost of modest macroeconomic and social contractions. Production scenarios consistently produce larger material savings than their FD counterparts for the same material, in proportion to the intermediate channel's dominance. Employment effects are negative (−0.01% to −0.02% LT) across all intermediate production scenarios except Sc10 (Metals), and wage shares are marginally compressed in the most intensive scenarios (Sc13). The production restructuring regime confirms the core IO insight: where intermediate demand dominates sector activation, A-matrix interventions are the primary CE instrument.

The CE Trilemma

The cross-cutting finding is a structural trilemma: ecological improvement trades off against macroeconomic stability and distributional equity in the intermediate production scenarios, while the construction scenarios produce a second, ecological-internal tension between material savings and emission rebound.

Scenario 4 vs Scenario 9 (Plastics) demonstrates that the trilemma can be partially bypassed through public procurement. For plastics — uniquely among all materials — government consumption (48.9%, 11.9 B EUR) rivals intermediate demand (44.0%, 10.7 B EUR) in absolute scale, making the two channels complementary rather than hierarchical. Neither scenario alone captures a majority of total plastics demand; effective plastics CE policy requires both instruments simultaneously.

Scenario 10 (Metals) is the most analytically anomalous outcome. It is the only production scenario showing expansionary macroeconomic effects ($Z1_{go} = +0.04\%$, $n = +0.03\%$ LT), driven by the lower A-matrix row-sum of secondary metals (0.238 vs 0.545): substituting secondary for primary frees EU intermediate resources economy-wide and generates a competitive dividend. Simultaneously, emissions increase (+0.27% LT) — the clearest structural rebound in the set. The mechanism is precisely identified: secondary metal production (EAF scrap-based steel) is more electricity-intensive upstream than primary production (blast furnace), and the high fossil emission intensity amplifies this into an EU emission increase even as material throughput falls strongly ($Z1_{mat} = -0.76\%$). Metals CE satisfies the economic arm of the trilemma while failing the ecological arm and requires complementary electricity decarbonisation.

Scenario 13 (Energy) represents maximum leverage and maximum asymmetry simultaneously. Ecological gains are the largest in the set ($emis = -1.56\%$, $x_{mat} = -1.27\%$ LT), and the current account improvement (+1.81%) reflects reduced EU fossil import dependency. But the scenario is distributionally regressive (wage share −0.08%, profit share +0.14%) and produces the largest cross-border burdens on RoW energy exporters ($Z2_{b_s}$ rises, $Z2_n = -0.01\%$). The land use implication is notable: despite saving the most emissions, EU land use increases slightly (+0.16% LT) because renewable electricity infrastructure (wind and solar) requires substantially more land per unit of output than centralised fossil power plants. EU energy CE is the highest-leverage domestic CE intervention while simultaneously the most asymmetric in its international consequences.

Construction scenarios (Sc5, Sc14) define a third internal tension: material use falls (−0.84% and −0.35% LT respectively) while emissions, land use, and water use all rebound sharply (+5.19% and +2.15%). Renovation/refurbishment is more energy-intensive per unit of output than new construction in the current A-matrix, meaning the circular reallocation displaces primary material throughput but increases upstream fossil energy demand. The ecological trilemma here is not social vs. ecological but rather material vs. atmospheric: circular construction saves physical material but worsens the emission, land, and water footprint unless accompanied by concurrent renovation energy standards and electricity decarbonisation.

Cross-Border Transmission: Material Flows and RoW Consequences

A key contribution of the MRIO-SFC framework is tracing bilateral demand flows between EU (Z1) and RoW (Z2), making the cross-border consequences of CE policy directly visible. The table below reports long-term RoW material use deviations:

RoW long-term % deviation from baseline — material use

	Sc 1	Sc2 1	Sc 3	Sc 4	Sc5 5	Sc6 1	Sc 7	Sc 8	Sc 9	Sc 10	Sc1 0	Sc1 1	Sc1 2	Sc1 3	Sc1 4
RoW total material use	0.0 1	-0.0 1	0	0.0 1	-0.0 5	-0.0 1	0	0	0.0 1	-0.1 5	0	0	-0.0 1	-0.0 2	
RoW primary material extraction	0.0 1	-0.0 3	0	0.0 0	-0.0 5	0.0 0	0	0	0.0 0	-0.1 0	0	0	-0.0 5	-0.0 2	

Four distinct cross-border transmission patterns emerge across the 14 scenarios.

Symmetric material contraction (most intermediate production scenarios). Scenarios 7–9, 11–12 produce broadly parallel reductions in material use in both Z1 and Z2, reflecting the global IO linkages through which EU production restructuring reduces demand for RoW intermediate inputs. The effects are small in absolute terms (Z2 mat near zero in most cases) but consistently negative: EU circular restructuring does not simply displace material throughput to the RoW — it reduces it globally. These scenarios are not unilaterally regressive on material grounds.

Metals (Sc10): the largest RoW material contraction. The most substantial cross-border material impact occurs in Scenario 10: Z2 total material use falls −0.15% LT and Z2 primary material extraction falls −0.10% LT. This is the largest RoW material reduction in the scenario set — larger, in absolute RoW terms, than any other production scenario. The mechanism is the competitive displacement of RoW metal production: as EU industry shifts to lower-input secondary metals, demand for RoW primary metal exports contracts, pulling down RoW extraction alongside EU extraction. The asymmetry is that RoW loses production and employment while EU macro-expands — a genuine welfare divergence mediated through the material trade channel.

Plastics and food scenarios: material leakage to RoW. Scenarios 4 (Gov Plastics) and 9 (Plastics Int) produce small positive RoW material effects (+0.01% mat), indicating partial production leakage: EU secondary plastics production has high import content, and the substitution generates demand for RoW inputs that partially offsets domestic primary material savings. Similarly, Scenario 1 (Food/Meat) produces a marginal RoW material increase (+0.01%), reflecting the displacement of some meat processing toward imported food products. These are the only scenarios in the set where RoW total material use increases — a leakage pattern that limits the global material savings of these CE interventions.

Energy scenarios: RoW primary extraction contraction. Scenarios 2 and 13 produce RoW primary material extraction contractions ($Z2 \times_{mat} = -0.03\%$ and -0.05% LT respectively), driven by the collapse in EU fossil fuel imports. This is the only transmission channel where a large RoW material reduction accompanies a large EU ecological gain. However, this contraction falls asymmetrically on fossil-fuel-exporting economies: the RoW aggregate obscures that specific exporting countries face much larger contractions than the aggregate suggests, with associated employment and fiscal consequences.

Construction (Sc5, Sc14): RoW material savings despite EU emission rebound. Despite the emission and land rebound in the EU, both construction scenarios produce small RoW material use reductions ($Z2 \text{ mat} = -0.05\%$ in Sc5, -0.02% in Sc14) because renovation uses less imported primary construction material than new build. The cross-border material effect is therefore mildly positive globally even when the EU domestic ecological balance is mixed.

Ecological Pressure Divergence: Land and Water

A secondary but analytically important finding is that emissions, land use, and water use diverge across scenarios. The three pressure dimensions are driven by different sector-specific intensity coefficients:

- Scenario 1 (Food/Meat): Emissions fall -0.42% but water use increases $+0.68\%$ EU — other food products (the secondary sector) are more water-intensive per unit of output than the primary meat sector. Land use falls only -0.12% , much less than emissions.
- Scenario 3 (Wood HH): Emissions are near zero but land use falls -0.36% — primary wood production is highly land-intensive; the circular shift releases land without changing fossil energy consumption.
- Scenario 7 (Wood Int): The largest land saving in the entire scenario set (EU land $= -2.30\%$ LT) from a scenario with near-zero emission effect (-0.01%). The land–emission decoupling is strongest here: forest-sector restructuring delivers a land dividend that the emission indicator entirely misses.
- Scenario 13 (Energy Int): Emissions fall -1.56% but land use increases $+0.16\%$ — renewable energy infrastructure requires more land per MWh than fossil plants. The energy scenario is simultaneously the best CE intervention for climate and the worst for land use.

- Construction (Sc5, Sc14): Emissions rebound (+5.19%, +2.15%) but land and water improve (−0.33%, −0.05% in Sc5). The circular reallocation reduces primary material and land extraction while worsening the atmospheric footprint.

These divergences reinforce that no single ecological indicator is sufficient for CE policy evaluation. Wood-sector restructuring is primarily a land-use intervention; energy-sector restructuring is primarily a climate intervention; food-sector restructuring is primarily an atmospheric and land intervention that worsens freshwater pressure.

Policy Implications

Four policy implications follow directly from the simulation evidence.

First, production restructuring requires macroeconomic stabilisation. Production scenarios consistently contract aggregate demand modestly, reduce wage shares, and expand profit shares. CE transitions in the production regime cannot be treated as costless structural adjustments — they require active macroeconomic management and labour market support.

Second, final-demand reallocation is primary for specific channels. For most material sectors where intermediate demand dominates (metals, wood, glass, cement), FD instruments engage structurally secondary channels and cannot substitute for A-matrix restructuring. However, for construction (investment-dominated), food (household-dominated), and plastics (government-procurement-dominated), FD instruments engage the primary demand channel directly and are first-order CE instruments in their own right.

Third, energy is the highest-leverage CE instrument, but land use is its shadow cost. Scenario 13 dominates all other production scenarios in both emission reduction (−1.56%) and primary material extraction reduction (−1.27%). However, renewable energy infrastructure requires substantially more land per unit of output than fossil plants, meaning that scaling this transition without land-use planning creates a new ecological trade-off that the emissions indicator does not capture.

Fourth, metals CE requires electricity decarbonisation as a complement, and its cross-border material dividend is real but regressive. Without simultaneous electricity decarbonisation, secondary metal production amplifies upstream emissions through the fossil energy link. With decarbonisation, Metals CE could simultaneously expand EU output, reduce global material throughput (including −0.15% in RoW), and reduce emissions — making it uniquely triple-positive. The current calibration shows only the first and second arm; the third requires a policy complement.

Conclusions